

UNIT 2



FUNDAMENTALS OF OPERATING SYSTEM



Q.1: DEFINE OPERATING SYSTEM. GIVE THREE EXAMPLES OF OPERATING SYSTEMS.**OPERATING SYSTEM (OS):**

An Operating System is a software which performs all the basic tasks like booting the computer, file management, memory management, process management, and controlling peripheral devices such as hard disk, printer, etc. It manages computer resources efficiently.

Examples: Most common operating systems are: DOS, IOS, Linux, Android, Mac OS and Windows.

Q.2: LIST THE FUNCTIONS OF AN OS. DESCRIBE THEM**FUNCTIONS OF OPERATING SYSTEM:**

Operating system manages every activity of a computer. It is the master control program that provides an interface for a user to communicate with computer. System software and application software run on operating system as shown in the given figure.

Operating System forms the following functions.

- (i) Booting
- (ii) Resource Management
- (iii) I/O Management
- (iv) File Management
- (v) Memory Management
- (vi) User Interface
- (vii) Process Management
- (viii) User Management

(i) BOOTING:

Booting is a process of starting the computer operating system. It checks the computer resources and makes it ready to perform different tasks.

(ii) Resource Management:

Operating system manages all the hardware and software resources. This includes allocation and de-allocation of processor, memory access to shared cache memory and access to network resources.

(iii) User Interface or Command Interpreter:

We interact with operating system through user interface. Command interpreter is one of the parts of operating system which reads the commands from user, interprets them and translates them into machine language instructions that the computer hardware can understand.

(iv) Memory Management:

Memory management module performs the task of allocation and de-allocation of memory space to programs and data in need of these resources.

Q.3: DEFINE INTERFACE AND USER INTERFACE. DESCRIBE THE TYPES OF USER INTERFACE.**INTERFACE:**

An interface is a program that allows a user to interact with the computer or another computer \ over a network.

User Interface:

A User Interface (UI) is the part of operating system, program, or devices that allows a user to enter and receive information. There are many types of User Interfaces. Two most common user interfaces are discussed below:

(i) Command Line Interface (CLI):

A Command Line Interface (CLI) is a screen or text based representation in which the user types the commands on place called prompt to operate the computer. Command contains string of characters. CLI is difficult to use because the user has to remember the commands and their syntaxes but it is fast in use because text mode takes less resources. It was primarily provided to users by computer terminals on UNIX and personal computers including MS-DOS and Apple DOS.

(ii) Graphical User Interface (GUI):

A GUI provides a user friendly environment where user can interact with computers through graphical objects such as menus, icons, buttons and other graphical objects. It is easy to use as users are supposed to just click on a picture to run commands without memorizing them. GUI is slower than CLI as graphical mode takes more memory and resources. Windows and IOS are the example of GUI.

Q.4: DESCRIBE THE TYPES OF OPERATING SYSTEM.

Types of Operating System:

The Operating System can be categorized as under:

- Single User and Multi-user OS
- Batch Processing OS
- Time Sharing OS
- Real Time Processing OS

(i) Single User And Multi-user Operating Systems:

In a Single User Operating System, a single user can access the computer system at time. These types of operating systems are commonly used. DOS for PCs and Windows 98 for PCs are example of single user operating system.

A Multi-user Operating System allows multiple users to access the computer at same operating system manages the memory and resources among the various users according to the requirement. Linux and UNIX are the most common examples, of the multi-user operating system.

(ii) Batch Processing Operating System:

General term of Batch Processing is used for programs that are executed with minimum human interaction. This type of operating system does not interact with the computer directly. Each user prepares his job and submits it to the computer operator. To speed up processing, jobs with similar needs are batched together and run as a group by an operator. The main function of a batch processing system is to automatically keep executing the jobs in a batch.

THE BENEFITS OF BATCH PROCESSING ARE:

- Batch systems can be shared by multiple users.
- The idle time for batch system is very less.
- Next job starts just after the current one.

(ii) Time Sharing Operating System:

Time sharing is a method that allows multiple users to share resources simultaneously. Multiple \ users can use specific computer at the same time in different places. A time sharing operating system is that in which from many tasks each task is given some time to execute so that all processes run smoothly without any problem. Multiple jobs are executed by the CPU by switching between them. As the system switches rapidly from one task or user to the other, a short time slot is given to each task or user for their executions and all feel that system is not shared. Thus, the user can receive an immediate response. For example, in a transaction processing, the processor executes each user program in a short time. These systems are also known as Multitasking Systems. The task can be from single user or from different users also. The time that each task gets to execute is called quantum. After this time interval is over OS switches over to next task.

(iv) Real Time Processing Operating System (RT OS):

A Real Time Processing Operating System is a time bound operating system which has fixed time limit. Processing has to be done within the defined time limit otherwise system will be failed. Real Time Process System classified into two parts:

Hard Real Time System:

A Hard Real Time System guarantees that critical tasks complete on time, even the shortest delay or urgency is not acceptable. Here system should meet the deadline. Missile systems and Air Traffic Control System are best examples of Hard Real Time System.

Soft Real Time System:

In this time constraint is less strict. The meeting of deadline is not compulsory for each task, every time. A critical real-time task gets priority over other tasks until it completes. It does not guarantee that task will be completed in defined time but before a certain time and according to the priority. Online Transaction System, Airline Ticket Reservation in which reservation could be delayed but should be done after checking available seats and completed before aeroplane flight, etc. are the examples of Soft Real Time System.

Q.5: IF YOU ARE A MANAGER OF A LARGE ORGANIZATION WHICH TYPE OF OS WILL YOU PREFER? JUSTIFY YOUR ANSWER WITH ANY TWO REASONS

If I am a manager of a large organization I will prefer Windows for operating system (OS).

Reason of Choosing Windows as OS:

Because Windows, in all its varieties, is so widely popular, we are less likely to have compatibility issues when sharing files with other organizations. The familiar interface means that there will be less time wasted on learning a new Windows also offers the

Q.6: WRITE DOWN THE MAJOR STEPS FOR INSTALLING WINDOWS OS.**INTERFACE:**

Major Steps for Installing Windows.

- (i) Insert the Windows Installation DVD/Flash Drive
- (ii) Restart your Computer
- (iii) Wait for the first startup screen
- (iv) Press or hold Del or F2 to enter the BIOS screen
- (v) Locate the Boot order/ Boot sequence
- (vi) Select Boot order/ Boot sequence as per your installation source
 - (a) USB Flash/USB Hard Disk
 - (b) USB CD/DVD ROM
 - (c) Internal CD/DVD ROM
- (vii) Select any option:
 - (a) Upgrade
 - (b) Customize Installation (Advance)
- (vii) Select any drive for installing your operating system.
- (ix) Follow the on-screen instructions.

Q.7: DEFINE THE PROCESS OF SOFTWARE INSTALLATION**SOFTWARE INSTALLATION:**

Software installation is the process of making programs ready for execution. Software installation or installer is a computer program that installs files, such as applications, drivers, or other software, onto a computer. Software is installed on to computer by various means. They can be downloaded from the internet. They can also be installed from an installation or DVD or from a USB flash drive.

Q.8 WHY IT IS IMPORTANT TO INSTALL AN ANTIVIRUS?

To keep our computer secure and free from viruses, we need to install an antivirus software. A number of free antivirus software are available online. AVG, Avast, Avira and Kasperaky are some of the free antivirus software.

These days we need to download installation file from respective antivirus software and run it but make sure that computer has the access of internet. After running the installation file, antivirus software will be automatically installed on our computer. Most important component of an antivirus software is the virus database which antivirus software updates time to time. We must update our virus database for preventing virus threats.

Q.9: WHAT IS THE PURPOSE OF OFFICE AUTOMATION SOFTWARE? WITH EXAMPLES, EXPLAIN HOW CAN OFFICE AUTOMATION HELP EMPLOYEES TO BE PRODUCTIVE AND EFFICIENT?**OFFICE AUTOMATION SOFTWARE:**

THE BENEFITS OF BATCH PROCESSING ARE:

Windows is an Operating System which needs to install other software to make the best use of our computer. Office Automation software is the most common software package that we install on our computers. Office Automation software is a group of computer programs that help users in their daily work. Generally you will find MS Office in school lab offices and other places.

Purpose of Office Automation:

Office automation eases organizational workloads by simplifying and automating processes like accounting, data management, training, facility management, and various administrative tasks. Office Automation Helps Employees to be Productive and Efficient: An office automation system doesn't consist of just going paperless. For example, maintaining all work product and client correspondence in a document management system. Yet, an office automation system offers organizations much more than going green by cutting down on paper. It is a powerful tool that can be used to eliminate manual processes, identify inefficient work flows, and facilitate informed decision making. Implementing an office automation system offers organizations a broad range of benefits. These benefits include: Improved accuracy Reduced time and resources Data insights and more informed decisions Business process improvement Reduced costs Data storage and management Improved Accuracy Humans make errors, properly implemented automated systems do not. Human errors, are not only inefficient in that they must be corrected and lead to productivity delays, but they can be costly. For instance, adding too many digits when paying an employee or vendor. Serious mistakes can lead to security and compliance issues, potentially fines and penalties. An office automation system limits human intervention in the transfer of data, which minimizes the occurrence of errors.

Reduced Costs:

By automating complex business processes, organizations don't need to invest as much into hiring for those tasks. As a result, operational costs are lower, while productivity and profit margins are significantly higher.

Reduced Time and Resources:

Through automation, organizations can accomplish more with less. By eliminating tedious and time-consuming processes, employees can spend their time on more high value tasks. For example, returning to the popular paperless benefit, with office automation software employees do not need to spend as much time collecting important documentation, entering that information, or filing away voluminous paper copies. The system does it for them.

Data Storage and Management:

Office automation systems simplify data storage while giving organizations the ability to monitor and control data through an electronic document management system. Common features include things like task management and reminder systems, as well as easy access to information by key stakeholders.

DATA INSIGHTS AND MORE INFORMED DECISIONS:

Office automation systems give organizations access to large data sets, reports, and analytics. Access to data enables more informed decisions. Moreover, by analyzing data and key performance indicators, organizations can implement improvements to their processes to remove bottlenecks and other inefficiencies. Through business process improvement, organizations optimize performance, improve the quality of their products or services, and ensure a higher level of compliance. The ability to improve business processes is what rates office automation systems from piecemeal automation technology.

Q.10: WRITE DOWN THE MAJOR STEPS FOR INSTALLING MS-OFFICE.**MAJOR STEPS FOR INSTALLING MS-OFFICE:**

To install MS-Office package we need to take following steps.

- (i)** Run the MS-Office setup from USB, DVD or Hard Disk backup.
 - (ii)** Check the box marked, accept the terms of this agreement and click on 'Continue'.
 - (iii)** Enter Product Key
 - (iv)** Click on Install Now or Customize procedure.
- Select the package which you want to install. Click on 'Install Now'.
- (vi)** Then installation begins.
 - (vii)** MS Office installer will notify automatically after finishing the installation.

Configuration Requirement For Installing MS-Office:

Before installing MS-Office we should make sure that hardware and software meet with the

minimum requirements for the version of MS-Office that we want to install. The minimum

requirements may differ from version to version. (For MS- Office 2013 the minimum requirements

are: 1.6 GHz processor, 1 GB RAM, 3 GB hard disk free space, minimum windows 7 operating system and graphics driver).